

DOE ASPECT · DE-FOA-0003646 · TEAMING PARTNER LIST TPL-0000073

Field by field, ready to paste

The DOE teaming form asks for nine fields. Each one is answered below in the form the portal expects. Paste them as written.

PORTAL FIELDS

eere-exchange.energy.gov, Teaming Partners

Investigator Name	Michael David Ward
Organization Name	PlastiBioFuel LLC
Organization Type	Small business. Service-disabled veteran-owned, SBA VetCert verified.
Topic Area	Chemicals from alternative and waste feedstocks. Post-consumer PET to terephthalic acid and downstream derivatives.
Website	plastibiofuel.com
Contact Address	12380 Iveson Drive, Haslet, TX 76052
Contact Email	michael@plastibiofuel.com
Contact Phone	817-319-7383

BACKGROUND AND CAPABILITIES

The one free-text field

WRITTEN TO A CHEMICALS PROGRAM, NOT AN AI PROGRAM

ASPECT scores feedstock, chemistry, and scale-up. The modeling program earns one sentence here because that is what this reader pays for. The full AI argument belongs in the DOE Genesis Mission pitch, where the topic asks for it.

PlastiBioFuel LLC is a service-disabled veteran-owned small business in Haslet, Texas, converting post-consumer PET into chemicals and fuel. The company works on an immobilized-enzyme route: a PET hydrolase held in a covalent organic framework, feeding a continuous spiral-flow reactor. What

is established today, in a signed final report from the University of North Texas covering January to June 2026. Free FAST-PETase breaks real post-consumer water bottle PET down to terephthalic acid, confirmed by HPLC against standards. The enzyme loads into a TAPB-BPDA-COF with no protein detectable in the supernatant at 280 nm. Immobilized turnover measured 122 micromol per hour per milligram at 24 hours, against a 246 to 348 literature range for the free enzyme, and the laboratory attributes that gap to mass transfer down the framework channel rather than to loss of enzyme activity. Three project milestones closed ahead of schedule. What is not yet built. The spiral-flow reactor exists as a validated model and not as hardware. The route from monomer to fuel-grade ethanol is the company's objective and has not been demonstrated. Nothing in this profile should be read as a pilot result. What PlastiBioFuel brings to a team. A waste feedstock with a measured conversion to a foundational molecule; framework and immobilization chemistry through a sponsored research agreement with Prof. Shengqian Ma at the University of North Texas; and a simulation and modeling program, built and run in-house, that decides which reactor conditions the laboratory tests next. What PlastiBioFuel is looking for. Partners with pre-pilot and pilot-scale reactor capacity, downstream separations, and offtake into chemical markets. The company is prepared to participate as a prime or as a subrecipient.

TWO STANDING RULES CARRIED INTO THIS TEXT

1. ARG Petro is offtake discussions and interest. There is no signed letter of intent and no executed offtake agreement, and no document in this package says otherwise.
2. Renewable fuel credit eligibility stays subject to final pathway qualification. This profile makes no credit claim at all.

PUBLICATION NOTICE

DOE publishes every field above on a list that any applicant can read. Submitting the request consents to that publication. Nothing here is confidential, and nothing here is business sensitive.